

Computer & Smartphone Diagnostics – Assignment Answers

1. Three Real-Life Uses of a Multimeter

PC power supply: A multimeter can measure PSU output voltage to check whether the power supply is providing the expected voltage.

Laptop charging problem: It can measure a charger's output voltage to help determine whether the adapter is supplying power correctly.

Smartphone charging circuit: A technician can check voltage and continuity around the charging connector and power circuit to locate a fault.

Safety: Use the correct meter setting and probes; internal electronics should be tested only by someone trained to work safely with them.

2. POST Card Video Demo – Brief Summary

A relevant YouTube demonstration is “**PC BIOS POST Testing/Diagnostics Card (ISA-PCI)**” by MausolfB. The video shows a POST card with a four-digit LED display that shows hexadecimal POST codes during BIOS startup. It also has LEDs for power rails and motherboard signals. The demonstration explains that a stuck RESET indicator can point to a reset-circuit short, faulty reset switch or wiring, motherboard connection problem, or BIOS fault. POST-code meanings vary by BIOS and motherboard, so the exact code should be checked against the relevant documentation.

YouTube reference: PC BIOS POST Testing/Diagnostics Card (ISA-PCI) —
<https://www.youtube.com/watch?v=ggoUkEjkwg>

Source verified from the YouTube result and video description.

3. Software Diagnostic Tool – Practical Activity

Recommended free tools include HWMonitor, Speccy, or CrystalDiskInfo. Run the selected tool on your own laptop or desktop and submit your own screenshot as practical evidence.

Step 1: Download the selected diagnostic tool from its official website.

Step 2: Install the software and launch it.

Step 3: Check the dashboard for CPU/GPU temperatures, fan speeds, clock speeds, memory information, or storage-health information supported by the tool.

Step 4: Take a screenshot of the main dashboard.

Step 5: Crop or blur personal information such as your Windows username, computer name, serial number, or other identifying details.

Step 6: Save the cleaned screenshot and attach it to the assignment.

4. Comparison of Diagnostic Tools

Tool	What it checks	Example Problem Detected
Multimeter	Electrical voltage, continuity, and power signals.	Incorrect PSU output voltage or a broken power connection.

Tool	What it checks	Example Problem Detected
POST card	BIOS Power-On Self-Test progress and hardware initialization codes.	A motherboard/BIOS or hardware initialization fault indicated by a stuck POST code.
Software diagnostics	Operating-system-visible temperatures, fan speeds, storage health, and other hardware information.	Overheating CPU or a storage drive reporting health/SMART warnings.

Conclusion: A multimeter is used for physical electrical measurements, a POST card helps diagnose early motherboard/BIOS startup stages, and software diagnostics monitor hardware information available after the operating system loads.